

Schur partitions of semi-structures

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Schur Ring

A **Schur Ring** is a partition S of a set G in which for all $B, C \in S$

$$BC = \sum_{D \in S} nD, n \in \mathbb{Z}$$

Elements of a Schur ring are called **primitive sets**.

Schur Ring Example

In $\mathbb{Z}_5$ we have the following Schur rings:

Trivial: $\{\{1\}, \{z, z^2, z^3, z^4\}\}$

Discrete: $\{\{1\}, \{z\}, \{z^2\}, \{z^3\}, \{z^4\}\}$

Inverses: $\{\{1\}, \{z, z^4\}, \{z^2, z^3\}\}$

We will verify that the inverse pairs do in fact form a Schur ring on this group.

Schur Ring Example

When verifying that a partition is a Schur ring, it can be helpful to consider it as a group ring.

So in $\mathbb{Z}_5$, the Schur ring $\{\{1\}, \{z, z^4\}, \{z^2, z^3\}\}$ behaves as follows:

$$1(z + z^4) = z + z^4 \quad 1(z^2 + z^3) = z^2 + z^3$$

$$(z + z^4)^2 = z^2 + 2(1) + z^3 = 2(1) + (z^2 + z^3)$$

$$(z^2 + z^3)^2 = z^4 + 2(1) + z = 2(1) + (z + z^4)$$

$$(z + z^4)(z^2 + z^3) = z^3 + z^4 + z + z^2 = (z^2 + z^3) + (z + z^4)$$

Semigroup

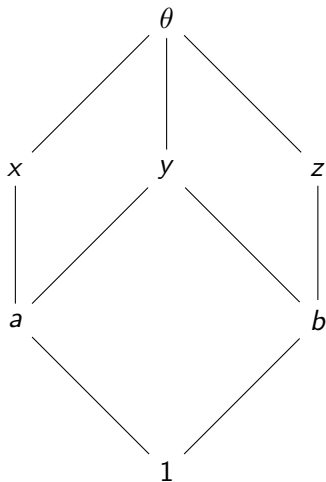
A **semigroup** G is a nonempty set with a binary operation defined on it that satisfies the following condition:

$$(1) \forall a, b, c \in G, (ab)c = a(bc) \text{ (Associative)}$$

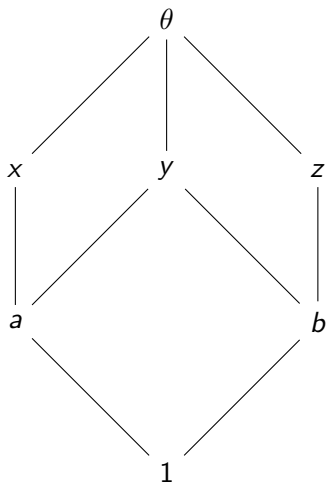
Furthermore, a semigroup that has an identity is called a **moniod**.

Semigroup Example

The diagram provides a semilattice, a subcategory of semigroup.



Semilattice Schur Ring Example



The Schur rings here are:

- 1) $\{\{1\}, \{a, b\}, \{x, z\}, \{y\}, \{\theta\}\}$
- 2) $\{\{1\}, \{a, b\}, \{x\}, \{y\}, \{z\}, \{\theta\}\}$
- 3) $\{\{1\}, \{a\}, \{b\}, \{x\}, \{y\}, \{z\}, \{\theta\}\}$

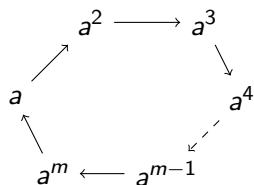
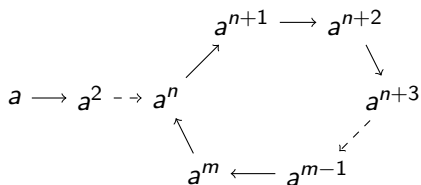
Monogenic Set

A set G is called **monogenic** if there is an element $g \in G$ where for all $h \in G$, h can be written as a power of g .

We quickly notice the similarity between monogenic sets and cyclic groups. It should be noted, however, that while cyclic groups necessarily have a group structure, a monogenic set need not, though it is associative, making all such sets monogenic semigroups. Both are always commutative.

Monogenic Semigroup

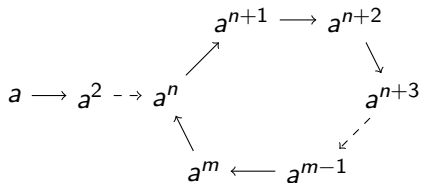
As all monogenic semigroups are generated by a single element, we can represent them with the following diagrams.



$$a \longrightarrow a^2 \longrightarrow a^3 \longrightarrow a^4 \longrightarrow a^5 \longrightarrow a^6 \dashrightarrow a^{m-1} \longrightarrow a^m$$

Cyclic subgroup

It will be noted that the loop here is not only a subsemigroup, but



is actually a cyclic group.

Schur Rings of Monogenic Semigroups

Let G be a semigroup, with $x \in G$. We say x is **decomposable** if $\exists a, b \in G$ where $ab = x$. Otherwise x is **indecomposable**.

Let S be a finite monogenic semigroup generated by a , where S is not a cyclic group. Since a is not in the cyclic subgroup, a is indecomposable.

Schur Rings of Monogenic Semigroups

With the set $S = \langle a \mid a^n = a^{n+m} \rangle$, let a be partitioned with a^k . Consider $(a + a^k + \dots)^k$, it is easy to see that regardless that value of k , a^k will result in this product. But as a is indecomposable, there is no product that can result in a . Which means that $\{a, a^k, \dots\}$ cannot be primitive set of any Schur ring of S .

So a must be partitioned on its own. But then so must every power of a , meaning the only Schur ring is the discrete one.

Final Result

Theorem

Every finite, non-cyclic, monogenic semigroup has exactly one Schur ring.